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# Tips for Linux users #64

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doegox opened on Oct 27, 2016

Last edited by doegox ▾

Contributor



As there is not really other place of discussion and my notes are probably not mature enough for inclusion in the official docs, I'm sharing a few tips here and hoping to gather yours as well!

I'm using a Debian, details may vary with other distros.

## Setup

First my `/etc/udev/rules.d/98-ChameleonMini.rules` to operate *and* flash the chameleon as regular user and stop being annoyed by pesky ModemManager:

```
# Rule for ChameleonMini RFID Research tool
ATTRS{product}=="Chameleon-Mini", SUBSYSTEMS=="usb", ATTRS{idVendor}=="16d0",
ATTRS{idProduct}=="04b2", GROUP="users", MODE="0666", SYMLINK+="chameleon",
ENV{ID_MM_DEVICE_IGNORE}="1"
# DFU mode
SUBSYSTEMS=="usb", ATTRS{idVendor}=="03eb", ATTRS{idProduct}=="2fde", GROUP="users", MODE="0666"
```



## Minicom

As terminal, minicom can do the job, but a few tweaks help a bit, with this file as `/etc/minicom/minirc.chameleon` or `~/.minirc.chameleon`:

```
pu port                /dev/chameleon
pu localecho           Yes
pu linewrap            Yes
# disable other transports than xmodem
pu pname1              YUNYY
pu pname2              YUNYY
pu pname4              NDNY
```



```
pu pname5      NDNYN
pu pname7      YUYNN
pu pname8      NDYNN
pu pname9      YUNYN
```

Then invoke it simply with:

```
minicom chameleon
```



xmodem transfers are possibles.

## Screen

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Usually I like `screen` as simple terminal but there is no easy way to activate local echo (see [this discussion](#))

## Socat

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So the alternative to get a simple terminal is to use socat.

A nice add-on is to use its readline support to get history, ctrl-r etc. Unfortunately Debian and Ubuntu versions compiled it without readline support, so if you want to have it you've to recompile the package by yourself (see [notes](#)) or use the original sources.

But getting history in your chameleon terminal is so great!

### cham

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```
#!/bin/bash
socat READLINE,history=$HOME/.chameleon_history /dev/chameleon,crlf
```



## File transfer: Download

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### cham-down

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```
#!/bin/bash
# Usage:
# $ cham-down [filename [slot_nr]]
# default is to dump current slot to dump.bin
[ ! -z $2 ] && echo -e "SETTING=$2\r" >/dev/chameleon && read foo </dev/chameleon
socat SYSTEM:"echo \"DOWNLOAD\r\" ;read foo; rx ${1:-dump.bin}" /dev/chameleon
```



# File transfer: Upload

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## cham-up

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```
#!/bin/bash
# Usage:
# cham-up [filename [slot_nr]]
# default is to upload dump.bin into current slot
[ -f "${1:-dump.bin}" ] || exit 1
[ ! -z $2 ] && echo -e "SETTING=$2\r" >/dev/chameleon && read foo </dev/chameleon
echo -e "UPLOAD\r" > /dev/chameleon
socat EXEC:"sx ${1:-dump.bin}" /dev/chameleon
```



## Flashing

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Here is my little script to compile, jump to bootloader, flash and jump back to normal mode in one go without having to unplug/plug the chameleon. This requires to install also the package `dfu-programmer`.

## cham-flash

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```
#!/bin/bash
cd /path/to/ChameleonMini/Firmware/Chameleon-Mini
make || exit 1
# Jump to bootloader mode
echo -e "\rUPGRADE\r" > /dev/chameleon && sleep 1
make program
# Jump to normal mode
dfu-programmer atxmega128a4u start
```



## Doc

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I also created a convenient

## cham-man

---

```
x-www-browser /path/to/ChameleonMini/Doc/Doxygen/html/index.html &
```



## Your turn

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Any other tips to share on your side?



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zenroth1752 on Oct 27, 2016



You are star - thank you



zenroth1752 on Feb 6, 2017



This tool allows the MFD dump to be in human readable format - <https://github.com/zhovner/mfdread>



**Peterthegreat** mentioned this [on Jul 30, 2017](#)

[setting & Dump Mifare Classic 4K #128](#)

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